

SIMATS ENGINEERING
ASSESSMENT TOOL 1

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COURSE NAME: Computer Networks
COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test – ANSWER KEY

1. ***Question:** 150 MB file transfer over 50 Mbps link; 1500-byte Transport segments; Data Link adds 40 bytes header+trailer per frame. Find segments, frames, DLL overhead, transmission time, and explain reliability.*

Answer:

Total data = 150 MB = 150,000,000 bytes.

Number of segments = $150,000,000 / 1500 = 100,000$ segments.

Number of frames = 100,000 (one frame per segment).

Data Link Layer overhead = $100,000 \times 40$ bytes = 4,000,000 bytes (4 MB).

Total data actually transmitted = $150,000,000 + 4,000,000 = 154,000,000$ bytes.

In bits = $154,000,000 \times 8 = 1,232,000,000$ bits.

Transmission time = $1,232,000,000 \text{ bits} / 50,000,000 \text{ bps} = 24.64$ seconds.

Reliability: The Transport Layer (e.g., TCP) provides segmentation, sequence numbers, acknowledgements, retransmission of lost segments, flow control and error checking (checksums) to guarantee end-to-end reliable delivery. The Data Link Layer provides framing, physical/MAC addressing and error detection (CRC) on each hop, and can request retransmission of corrupted frames, ensuring reliable node-to-node transfer.

2. Question: *Sliding window size = 6 frames, 1024 bytes/frame, sender transmits 48 frames; one frame lost per window. Find number of windows, retransmissions, and explain flow control.*

Answer:

Number of windows = $48 \text{ frames} / 6 \text{ frames per window} = 8$ windows.

Since 1 frame is lost in every window, retransmissions = $1 \times 8 = 8$ retransmissions.

Flow control: Sliding window flow control lets the sender transmit multiple frames (up to the window size) before waiting for an acknowledgement, instead of stopping after every single frame. This keeps the link utilised continuously, improves throughput, and still prevents the sender from overwhelming a slower receiver's buffer, giving a good balance between efficiency and reliability.

3. Question: 12,000-byte IP packet fragmented at a router with MTU = 1500 bytes; each fragment carries a 20-byte IP header. Find number of fragments, header overhead, and explain the Network Layer's role.

Answer:

Usable payload per fragment = MTU – header = 1500 – 20 = 1480 bytes.

Number of fragments = $\lceil 12,000 / 1480 \rceil = \lceil 8.11 \rceil = 9$ fragments.

Total header overhead = 9×20 bytes = 180 bytes.

Total data transmitted (with headers) $\approx 12,000 + 180 = 12,180$ bytes.

Network Layer's role: The Network Layer (IP) fragments large packets to fit the MTU of each link, assigning each fragment an identification number, fragment offset and a more-fragments flag. At the destination, the Network Layer reassembles the fragments in the correct order using these fields before passing the complete packet up to the Transport Layer, ensuring successful end-to-end delivery even across links with different MTU sizes.

4. Question: Sliding window size = 6 frames, 1024 bytes/frame, sender transmits 48 frames; one frame lost per window. Find number of windows, retransmissions, and explain flow control. (Same as Q2)

Answer:

Number of windows = $48 / 6 = 8$ windows.

Retransmissions = 1 lost frame per window $\times 8$ windows = 8 retransmissions.

Flow control: By allowing several frames to be sent before an acknowledgement is required, the sliding window mechanism increases link utilisation and reduces idle waiting time, while still regulating the sender's rate to match the receiver's processing/buffer capacity — improving overall communication efficiency without causing data loss due to buffer overflow.

5. Question: 600 MB video; Session Layer checkpoint every 60 MB; failure occurs after 390 MB transferred. Find checkpoints before failure and data to be retransmitted; explain importance of synchronization.

Answer:

Checkpoints occur at 60, 120, 180, 240, 300 and 360 MB → 6 checkpoints completed before failure (since $390/60 = 6.5$, only 6 full checkpoints are reached).

Data to be retransmitted after recovery = data sent after the last checkpoint = $390 - 360 = 30$ MB.

Importance of synchronization: The Session Layer inserts synchronization checkpoints so that if a session fails, transmission can resume from the last checkpoint instead of restarting from the beginning. This minimises retransmitted data, saves time and bandwidth, and makes long data transfers (like large media files) resilient to interruptions.

6. Question: *Presentation Layer compresses a 900 MB file by 40%; encryption adds 5% overhead to the compressed size. Find compressed size, encrypted size, total % reduction, and explain the Presentation Layer's role.*

Answer:

Compressed size = $900 \text{ MB} \times (1 - 0.40) = 540 \text{ MB}$.

Encrypted size = $540 \text{ MB} \times (1 + 0.05) = 567 \text{ MB}$.

Total reduction compared to original = $(900 - 567) / 900 \times 100 = 37\%$.

Presentation Layer's role: It compresses data to reduce the volume transmitted (saving bandwidth and time) and encrypts it to protect confidentiality and integrity during transmission. It also handles data translation/formatting (e.g., encoding) so data from different systems can be understood uniformly, improving both efficiency and security of communication.

7. Question: *FTP transfers 3 GB over an effective throughput of 120 Mbps; Application Layer divides data into 3 MB requests. Find total requests, transmission time, and discuss Application Layer reliability.*

Answer:

Total data = 3 GB = 3000 MB.

Number of requests = $3000 \text{ MB} / 3 \text{ MB} = 1000$ requests.

Total data in bits = $3,000,000,000 \text{ bytes} \times 8 = 24,000,000,000$ bits.

Transmission time = $24,000,000,000 \text{ bits} / 120,000,000 \text{ bps} = 200$ seconds.

Application Layer reliability: The Application Layer (e.g., FTP) provides the user-facing service and structures data into logical requests/responses, but relies on the underlying Transport Layer (TCP) for guaranteed, in-order, error-checked delivery. Together they provide reliable session establishment, authentication, request tracking and error reporting to the user.

8. Question: *Processing delays: Application 4 ms, Transport 3 ms, Network 5 ms, Data Link 2 ms, Physical 1 ms; propagation delay 18 ms; transmission delay 12 ms. Find total end-to-end delay and explain each layer's contribution.*

Answer:

Total processing delay = $4 + 3 + 5 + 2 + 1 = 15$ ms.

Total end-to-end delay = processing delay + propagation delay + transmission delay = $15 + 18 + 12 = 45$ ms.

Layer contributions: The Application Layer prepares/formats the request; the Transport Layer segments data and manages reliability; the Network Layer performs routing and addressing decisions; the Data Link Layer frames the data and handles link-level error control; the Physical Layer converts data to signals for the medium. Propagation delay depends on distance and medium, and transmission delay depends on packet size and link bandwidth — together all these components add up to the total end-to-end delay experienced by the packet.

9. Question: 80 MB useful data sent using 1600-byte frames, each frame carrying 80 bytes of protocol overhead. Find number of frames, total overhead, transmission efficiency, and explain the effect of overhead on performance.

Answer:

Useful payload per frame = $1600 - 80 = 1520$ bytes.

Number of frames = $\lceil 80,000,000 / 1520 \rceil = 52,632$ frames (approx).

Total overhead = $52,632 \times 80$ bytes $\approx 4,210,560$ bytes (≈ 4.21 MB).

Transmission efficiency = $\text{payload/frame size} \times 100 = 1520/1600 \times 100 = 95\%$.

Effect of overhead: Protocol overhead (headers/trailers added by each layer) consumes part of every frame's capacity without carrying user data, lowering the effective/goodput throughput of the network. Higher overhead relative to frame size reduces efficiency, so choosing an appropriately large frame size (within the MTU) helps improve overall network performance.

10. Question: 240 MB medical record: Presentation Layer compresses by 30%, Transport Layer segments into 1400-byte units, Data Link Layer adds 60 bytes overhead per frame. Find compressed size, segments, overhead, final data over the Physical Layer, and explain layer cooperation.

Answer:

Compressed size = $240 \text{ MB} \times (1 - 0.30) = 168 \text{ MB} = 168,000,000$ bytes.

Number of segments = $168,000,000 / 1400 = 120,000$ segments.

Data Link Layer overhead = $120,000 \times 60$ bytes = $7,200,000$ bytes (7.2 MB).

Final data transmitted across the Physical Layer = $168,000,000 + 7,200,000 = 175,200,000$ bytes (175.2 MB).

Layer cooperation: The Presentation Layer compresses (and can encrypt) the sensitive medical data for efficient, secure transfer; the

Transport Layer breaks the compressed data into manageable segments and ensures reliable, ordered delivery; the Data Link Layer frames each segment with error-control overhead for safe transfer across each physical link; and the Physical Layer transmits the resulting bits over the IoT network medium. Their combined operation ensures the patient record is transmitted securely, reliably and efficiently end to end.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand